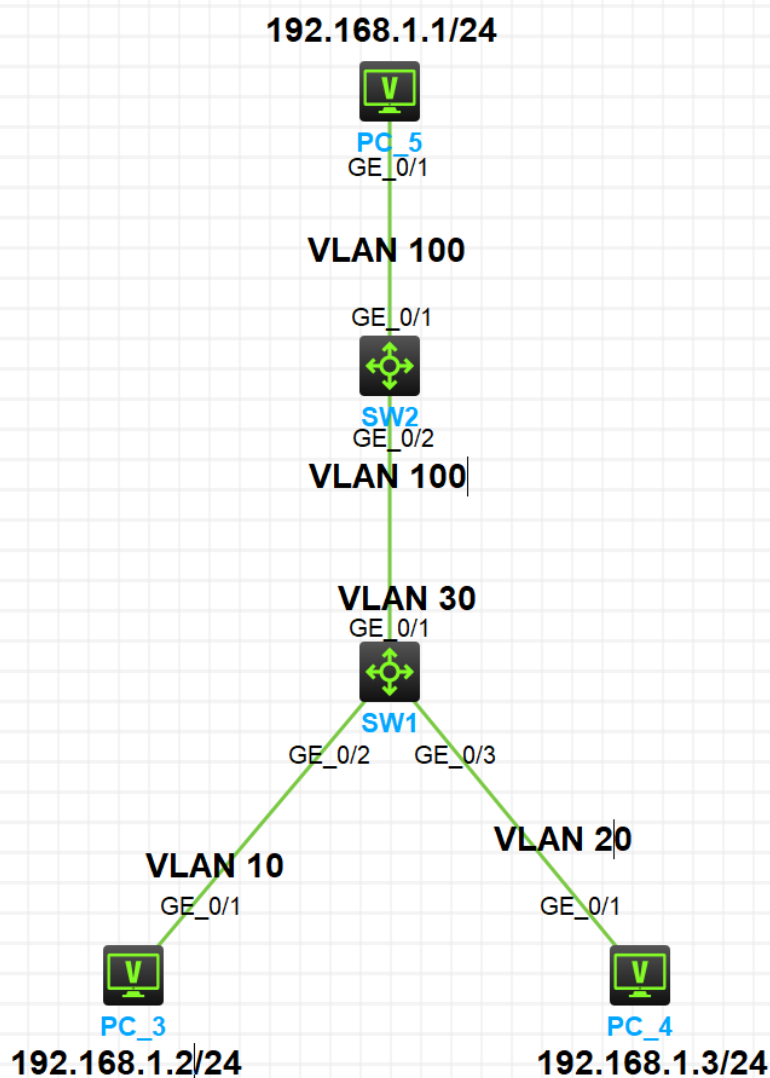


私有VLAN实验

实验拓扑

实验需求

- 1.按照图标配置 IP 地址
- 2.在 SW1 上配置 Private VLAN，Primary VLAN 为 Vlan30，Secondary VLAN 为 Vlan10 和 Vlan20
- 3.SW2通过Vlan100下行连接SW1，要求PC3和PC4都能以Vlan100访问PC5



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- 1.按照图标配置 IP 地址
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实验配置

```
1 [SW2]vlan 100
2 [SW2-vlan100]port GigabitEthernet 1/0/1
3 [SW2-vlan100]port GigabitEthernet 1/0/2
4 ---
5 [SW1]vlan 10
6 [SW1-vlan10]vlan 20
7 [SW1-vlan20]vlan 30
8 [SW1-vlan30]private-vlan primary
9 [SW1-vlan30]Nov 11 00:01:24:133 2025 SW1 LLDP/5/LLDP_PVID_INCONSISTENT: PVID mismatch
10 discovered on GigabitEthernet1/0/1 (PVID 1), with SW2 GigabitEthernet1/0/2 (PVID 100).
11 [SW1-vlan30]private-vlan secondary 10 20
12 [SW1-vlan30]int g1/0/1
13 [SW1-GigabitEthernet1/0/1]port private-vlan 30 promiscuous
14 [SW1-GigabitEthernet1/0/1]int g1/0/2
15 [SW1-GigabitEthernet1/0/2]port access vlan 10
16 [SW1-GigabitEthernet1/0/2]port private-vlan host
17 [SW1-GigabitEthernet1/0/2]int g1/0/3
18 [SW1-GigabitEthernet1/0/3]port private-vlan host
19 [SW1-GigabitEthernet1/0/3]port access vlan 20
20 ---
21
```

结果验证

1.连通性测试

```
1 <H3C>ping 192.168.1.1
2 Ping 192.168.1.1 (192.168.1.1): 56 data bytes, press CTRL_C to break
3 56 bytes from 192.168.1.1: icmp_seq=0 ttl=255 time=1.830 ms
4 56 bytes from 192.168.1.1: icmp_seq=1 ttl=255 time=1.405 ms
5 ---
6 <H3C>ping 192.168.1.1
7 Ping 192.168.1.1 (192.168.1.1): 56 data bytes, press CTRL_C to break
8 56 bytes from 192.168.1.1: icmp_seq=0 ttl=255 time=1.866 ms
9 56 bytes from 192.168.1.1: icmp_seq=1 ttl=255 time=1.167 ms
10 ---
11 <H3C>ping 192.168.1.2
12 Ping 192.168.1.2 (192.168.1.2): 56 data bytes, press CTRL_C to break
13 56 bytes from 192.168.1.2: icmp_seq=0 ttl=255 time=1.114 ms
14 56 bytes from 192.168.1.2: icmp_seq=1 ttl=255 time=1.124 ms
15
16 <H3C>ping 192.168.1.3
```

```
17 Ping 192.168.1.3 (192.168.1.3): 56 data bytes, press CTRL_C to break
18 56 bytes from 192.168.1.3: icmp_seq=0 ttl=255 time=1.057 ms
19 56 bytes from 192.168.1.3: icmp_seq=1 ttl=255 time=1.137 ms
20
```

2.查看Private vlan 信息

```
1 [SW1]display private-vlan 30
2 Primary VLAN ID: 30
3 Secondary VLAN ID: 10, 20
4
5 VLAN ID: 30
6 VLAN type: Static
7 Private VLAN type: Primary
8 Route interface: Not configured
9 Description: VLAN 0030
10 Name: VLAN 0030
11 Tagged ports: None
12 Untagged ports:
13     GigabitEthernet1/0/1          GigabitEthernet1/0/2
14     GigabitEthernet1/0/3
15
16 VLAN ID: 10
17 VLAN type: Static
18 Private VLAN type: Secondary
19 Route interface: Not configured
20 Description: VLAN 0010
21 Name: VLAN 0010
22 Tagged ports: None
23 Untagged ports:
24     GigabitEthernet1/0/1          GigabitEthernet1/0/2
25
26 VLAN ID: 20
27 VLAN type: Static
28 Private VLAN type: Secondary
29 Route interface: Not configured
30 Description: VLAN 0020
31 Name: VLAN 0020
32 Tagged ports: None
33 Untagged ports:
34     GigabitEthernet1/0/1          GigabitEthernet1/0/3
35 ---
36 [SW2-vlan100]display vlan 100
37 VLAN ID: 100
38 VLAN type: Static
39 Route interface: Not configured
40 Description: VLAN 0100
41 Name: VLAN 0100
42 Tagged ports: None
43 Untagged ports:
44     GigabitEthernet1/0/1          GigabitEthernet1/0/2
```

